P0 Walk-Forward Validation

Point-in-time backtest of price-derived factor blocks | 2020-07-01 to 2025-01-31 | monthly rebalance, 52 dates | generated 2026-08-20T15:26:20

What this report does and does not establish

This is not a validation of Model 5.0. It exercises only the price-derived blocks. Quality, growth and value together carry 70% of the production score and require point-in-time fundamentals that are not yet ingested, so none of them are tested here.

It answers one question from p0.md §7E: **how much of the model's behaviour is momentum?** A momentum-only ranking that performs close to the full model would mean the fundamental blocks are decoration. Read every number below as evidence about momentum and risk, not about Model 5.0.

Two further gaps: these ablations rank **market-wide** where production ranks within sector, and momentum-only omits RS_Sector_6M_Pct (20% of the production momentum block) because the bhavcopy archive carries no sector classification.

Integrity controls

Control	Status
Universe source	Archived NSE bhavcopy per session (point-in-time, includes delisted names)
Survivorship	3380 securities; 518 delisted retained in-sample
Identifier	ISIN-bridged Security_ID; 416 face-value changes linked
Execution	Signal on close of t, fill at open of next confirmed session
Corporate actions	659 split/bonus adjusted, 11580 dividends, 620 unadjustable events excluded
Delisting policy	haircut@0.50
Costs	effective-dated Indian charges + 10bp half-spread + sqrt impact at Rs 1.00 lakh/position
Eligibility	min turnover Rs 20.00 lakh/day, trading frequency >= 0.8, history >= 200 sessions, INE-class only (ETFs excluded)
Parameters	Frozen; no re-optimisation across the window

Eligible universe per rebalance

Metric	Value
Rebalance dates	52
Mean eligible	1353
Min / max eligible	850 / 1682
Mean considered before filters	1817

Rank information coefficient — Gross of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	52	0.0078	0.0090	60%	-0.1471	0.1777
growth only	3M	52	0.0040	-0.0001	50%	-0.1362	0.1112
growth only	6M	52	0.0017	0.0017	52%	-0.1060	0.1207
model 5	1M	52	0.0406	0.0358	63%	-0.2904	0.3704
model 5	3M	52	0.0486	0.0508	77%	-0.2410	0.3229
model 5	6M	52	0.0631	0.0623	75%	-0.2862	0.2683
momentum only	1M	52	0.0268	0.0223	60%	-0.3134	0.3183
momentum only	3M	52	0.0464	0.0542	75%	-0.1838	0.2399
momentum only	6M	52	0.0711	0.0916	77%	-0.3010	0.2626
quality only	1M	52	0.0174	0.0101	52%	-0.2126	0.2553
quality only	3M	52	0.0089	0.0083	56%	-0.2169	0.2387
quality only	6M	52	0.0061	0.0147	54%	-0.2188	0.1837
random ranking	1M	52	-0.0011	-0.0032	46%	-0.0833	0.0651
random ranking	3M	52	0.0019	0.0050	56%	-0.0640	0.0591
random ranking	6M	52	0.0047	0.0046	50%	-0.0550	0.0503
simple quality momentum	1M	52	0.0310	0.0277	56%	-0.3133	0.3621
simple quality momentum	3M	52	0.0429	0.0474	71%	-0.2353	0.2644
simple quality momentum	6M	52	0.0647	0.0756	81%	-0.3046	0.2451
value only	1M	52	0.0354	0.0402	73%	-0.0887	0.1464
value only	3M	52	0.0515	0.0459	87%	-0.1053	0.1871
value only	6M	52	0.0612	0.0558	88%	-0.0869	0.1956

Portfolio and bucket evidence — Gross of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	3.34%	3.34%	+0.00%	0.612	+0.71%
equal weight universe	3M	9.41%	9.41%	+0.00%	0.709	+1.93%
equal weight universe	6M	16.61%	16.61%	+0.00%	0.345	+3.13%
growth only	1M	3.34%	3.79%	+0.45%	-0.091	-0.14%
growth only	3M	9.41%	9.92%	+0.51%	-0.164	-0.74%

growth only	6M	16.61%	16.73%	+0.13%	-0.115	-1.63%
model 5	1M	3.34%	4.84%	+1.51%	0.455	+0.28%
model 5	3M	9.41%	12.38%	+2.97%	0.612	+0.42%
model 5	6M	16.61%	20.59%	+3.98%	0.867	+2.81%
momentum only	1M	3.34%	4.38%	+1.05%	0.879	+1.49%
momentum only	3M	9.41%	12.72%	+3.31%	0.891	+4.83%
momentum only	6M	16.61%	23.18%	+6.58%	0.988	+10.71%
quality only	1M	3.34%	2.69%	-0.65%	-0.891	-0.91%
quality only	3M	9.41%	8.21%	-1.21%	-0.879	-2.78%
quality only	6M	16.61%	15.05%	-1.56%	-0.915	-4.88%
random ranking	1M	3.34%	3.33%	-0.01%	0.139	+0.09%
random ranking	3M	9.41%	10.02%	+0.61%	0.018	+0.10%
random ranking	6M	16.61%	17.45%	+0.85%	0.321	+0.50%
simple quality momentum	1M	3.34%	3.15%	-0.19%	0.879	+0.39%
simple quality momentum	3M	9.41%	8.08%	-1.33%	0.842	+1.14%
simple quality momentum	6M	16.61%	14.66%	-1.94%	0.903	+4.59%
value only	1M	3.34%	6.32%	+2.98%	0.551	+0.85%
value only	3M	9.41%	18.61%	+9.20%	0.600	+2.34%
value only	6M	16.61%	37.31%	+20.71%	0.648	+5.76%

Rank information coefficient — Net of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	52	0.0194	0.0229	63%	-0.1326	0.1831
growth only	3M	52	0.0108	0.0079	58%	-0.1300	0.1168
growth only	6M	52	0.0063	0.0068	56%	-0.1015	0.1285
model 5	1M	52	0.0659	0.0595	77%	-0.2677	0.3925
model 5	3M	52	0.0632	0.0681	79%	-0.2309	0.3321
model 5	6M	52	0.0732	0.0761	79%	-0.2792	0.2832
momentum only	1M	52	0.0411	0.0344	65%	-0.2958	0.3304
momentum only	3M	52	0.0549	0.0617	75%	-0.1770	0.2465
momentum only	6M	52	0.0767	0.0998	81%	-0.2955	0.2703
quality only	1M	52	0.0420	0.0360	67%	-0.1946	0.2697
quality only	3M	52	0.0229	0.0211	62%	-0.2049	0.2489

quality only	6M	52	0.0159	0.0247	56%	-0.2107	0.1976
random ranking	1M	52	-0.0009	-0.0017	46%	-0.0808	0.0664
random ranking	3M	52	0.0020	0.0048	56%	-0.0620	0.0596
random ranking	6M	52	0.0048	0.0051	50%	-0.0546	0.0526
simple quality momentum	1M	52	0.0537	0.0434	63%	-0.2917	0.3827
simple quality momentum	3M	52	0.0562	0.0637	77%	-0.2272	0.2725
simple quality momentum	6M	52	0.0737	0.0794	81%	-0.2979	0.2599
value only	1M	52	0.0333	0.0399	73%	-0.0839	0.1379
value only	3M	52	0.0502	0.0424	87%	-0.1012	0.1843
value only	6M	52	0.0604	0.0558	88%	-0.0826	0.1927

Portfolio and bucket evidence — Net of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	1.54%	1.54%	+0.00%	0.673	+0.66%
equal weight universe	3M	7.62%	7.62%	+0.00%	0.709	+1.89%
equal weight universe	6M	14.81%	14.81%	+0.00%	0.515	+3.08%
growth only	1M	1.54%	2.25%	+0.70%	0.624	+0.28%
growth only	3M	7.62%	8.38%	+0.76%	0.030	-0.32%
growth only	6M	14.81%	15.19%	+0.37%	-0.018	-1.21%
model 5	1M	1.54%	3.53%	+1.99%	0.770	+1.28%
model 5	3M	7.62%	11.07%	+3.45%	0.770	+1.42%
model 5	6M	14.81%	19.27%	+4.46%	0.939	+3.81%
momentum only	1M	1.54%	2.66%	+1.11%	0.964	+1.92%
momentum only	3M	7.62%	11.00%	+3.38%	0.915	+5.25%
momentum only	6M	14.81%	21.46%	+6.65%	0.988	+11.13%
quality only	1M	1.54%	1.49%	-0.06%	-0.285	+0.01%
quality only	3M	7.62%	7.00%	-0.62%	-0.867	-1.86%
quality only	6M	14.81%	13.84%	-0.98%	-0.879	-3.96%
random ranking	1M	1.54%	1.62%	+0.07%	0.139	+0.09%
random ranking	3M	7.62%	8.30%	+0.68%	0.042	+0.11%
random ranking	6M	14.81%	15.74%	+0.93%	0.442	+0.50%
simple quality momentum	1M	1.54%	1.88%	+0.33%	0.988	+1.19%
simple quality momentum	3M	7.62%	6.81%	-0.81%	0.855	+1.95%
simple quality momentum	6M	14.81%	13.39%	-1.42%	0.915	+5.40%
value only	1M	1.54%	4.38%	+2.84%	0.648	+1.25%
value only	3M	7.62%	16.67%	+9.05%	0.600	+2.74%
value only	6M	14.81%	35.38%	+20.57%	0.636	+6.16%

Turnover

One-way turnover of a top-20 portfolio per rebalance. 1.0 means the portfolio was fully replaced. High turnover is what converts a real gross edge into no net edge.

Strategy	Mean one-way turnover	Rebalances
equal weight universe	0.055	52
growth only	0.122	52
model 5	0.361	52
momentum only	0.486	52
quality only	0.086	52
random ranking	0.973	52
simple quality momentum	0.419	52
value only	0.173	52

Fill coverage

Share of scored positions that produced a usable forward return. A dropped position is recorded with a reason rather than silently omitted; a thin horizon here weakens every statistic above it.

Horizon	Coverage	Positions	Main reasons dropped
1M	99.6%	535048	unadjustable_action 2016, no_entry_price 56
3M	98.9%	535048	unadjustable_action 5792, no_entry_price 56
6M	97.9%	535048	unadjustable_action 11168, no_entry_price 56

How to read this

The random_ranking row is the calibration. It is a seeded random score, so its IC is what "no signal" measures on this universe and horizon. Any strategy whose IC is not clearly separated from it has not demonstrated anything.

equal_weight_universe scores every name identically, so its rank IC is undefined rather than zero. It is present as the return benchmark from p0.md §7B, not as a ranking.

Judge consistency before magnitude. A mean IC of 0.03 with 70% of periods positive is stronger evidence than a mean of 0.06 driven by two periods. Check the %+ and worst columns before the mean.

Net is the number that matters. If a gross edge disappears net of costs, the ranking may still be real but is not tradable at this position size and turnover.

Known limitations of this run

Restatement bias remains. No fundamentals are used here, so it does not bite yet — but it will once the XBRL panel lands, for any period falling back to current values rather than as-filed ones.

Delisting recovery is an assumption, not an observation. The bhavcopy does not record why a security stopped trading. Re-run with --delisting-strategy zero and last_close to bound how much the headline depends on it.

Period count is modest. Monthly rebalancing over roughly four years is enough to judge cross-sectional rank IC and its stability. It is not enough to claim regime robustness across a full cycle.

Overlapping horizons are not independent. A 6M horizon sampled monthly reuses most of its window, so any t-statistic on per-period ICs overstates significance and is reported as a rough guide only.

Cost assumptions are documented estimates. Brokerage defaults to zero (discount-broker delivery); half-spread and impact coefficient are modelled, not measured. Gross and net are shown separately for that reason.

Next step

Phase B: ingest NSE XBRL filings into a point-in-time fundamental panel keyed on filing timestamp, which unlocks the quality, growth and value blocks and makes a genuine Model 5.0 walk-forward possible. Until then the fundamental 70% of the score remains untested.